CHAPTER 1

SIGNALS AND SYSTEMS: FUNDAMENTAL CONCEPTS

In this chapter a mathematical representation for signals will be introduced. Some of the fundamental system concepts will be defined.

1.1. SIGNALS

We use the word "signal" to mean the variation of a physical -and sometimes nonphysical- quantity with respect to one or more independent variable. A speech signal, for example, is represented by the variation of acoustic pressure with respect to time, whereas a black and white picture is represented as brightness function of two spatial variables. Here we will mostly consider signals that are functions of a single independent variable. For convenience we will mostly refer to the independent variable as "time", although it may represent something else in a particular application,

It is possible to represent signals in different forms but in all cases the information contained by the signal is in the pattern of the signal.

Signals are basically in one of two forms: continuous-time signals and discrete-time signals. A continuous-time signal is defined in the continuous variation of the independent variable, time. The signal itself may have discontinuities, but the independent variable is necessarily continuous. Discrete-time signals on the other hand are defined only for specified time instants, i.e. the independent variable can take discrete values only. Note that, apart from these values the signal is not zero; or anything else, it is simply not defined. The speech signal is a continuous time signal whereas, the speech signal sampled to be processed on a computer or foreign currency exchange rates announced daily are discrete time signals.

To distinguish between the two types, the symbol ***t*** will be used for the continuous time independent variable in parentheses (.) e.g. $x(t)$. For the discrete time case the independent variable ***n*** will be shown in square brackets [.], e.g. $x[n]$. Graphical representations for continuous-time signal $x(t)$ and discrete time signal $x[n]$ are shown in Figure l.l.l.

Since $x[n]$ is defined only for integer values of ***n***, it is often referred to as a discrete-time *sequence*.

A discrete time signal may represent a phenomenon for which the independent variable is inherently discrete (e.g. blood pressures of different patients examined in a hospital) or it may represent successive "samples" of a continuous time process (e.g. .blood pressure of a certain patient taken at regular intervals or speech signals sampled for computer processing).

In this text, we will treat continuous-time and discrete-time signals separately but in parallel to see similarities and differences.

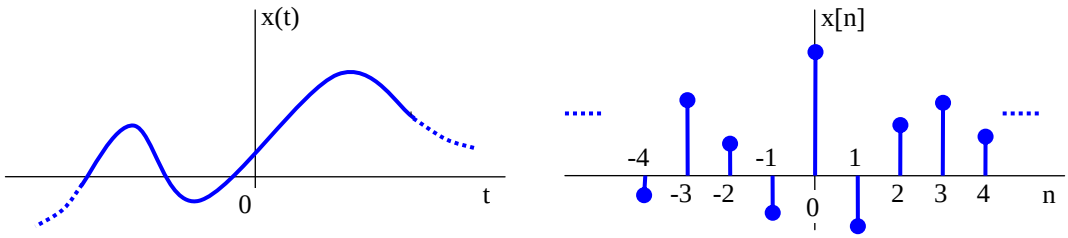

Fig.l.l.l. Graphical representations of (a) continuous-time and (b) discrete-time signals.

1.2. TRANSFORMATIONS OF THE INDEPENDENT VARIABLE

Consideration of signals related by simple modifications of the independent variable is important since, in addition to representing certain phenomena, they are often useful in investigating the properties of a certain signal.

The signal $x[-n]$ can be obtained by reflecting $x[n]$ with respect to the vertical axis, as shown in Figure 1.2.1. $x(-t)$ can be obtained from $x(t)$ in a similar manner. Just to help visualize the phenomena, note that, if $x(t)$ is recorded on a magnetic tape, this transformation corresponds to playing it backwards.

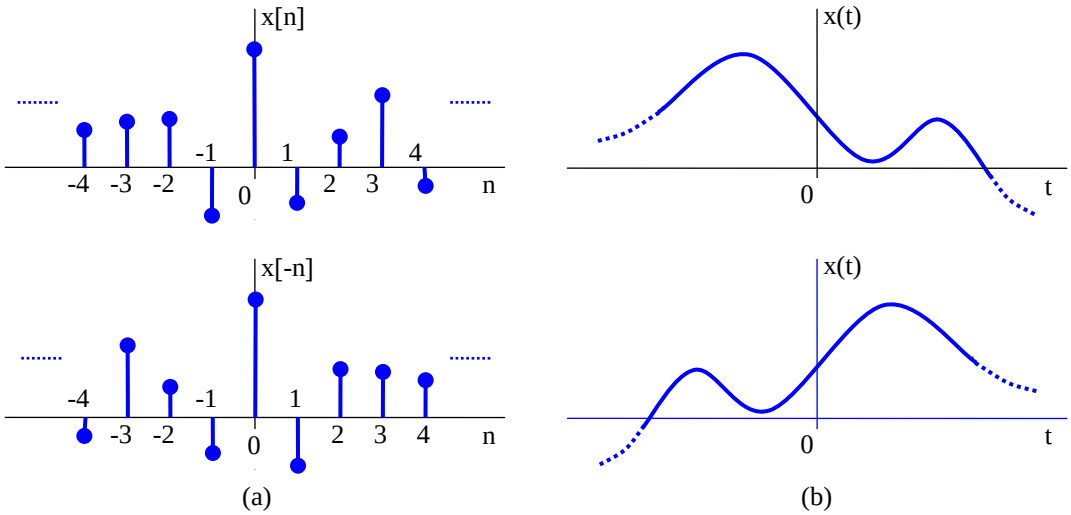

Fig.1.2.1. (a) A discrete-time and (b) a continuous-time signal and their reflections about the vertical axis.

Another simple transformation of the independent variable is multiplication; or scaling by a constant e.g. $x(at)$ as shown in Figure 1.2.2. For the magnetic tape example, this corresponds to playing the tape faster or slower.

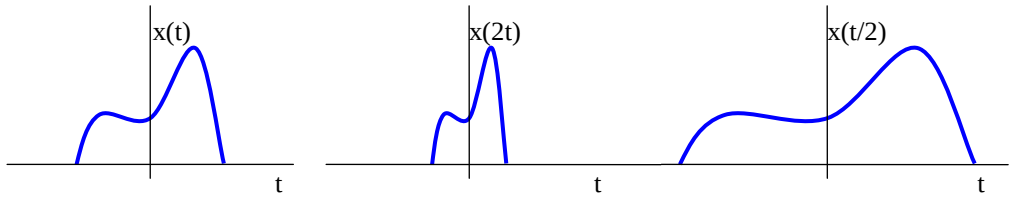

Fig.1.2.2. Continuous time signals related by time scaling.

Note that this transformation is rather ambiguous for discrete time systems, for the signal either becomes partially undefined or some of the information is lost.

A third simple transformation is obtaining $x[n-n_0]$ or $x(t-t_0)$ by shifting $x[n]$ or $x(t)$ to the left (if $\boldsymbol{n}_o$ or $\boldsymbol{t}_o$ is negative) or to the right (if $\boldsymbol{n}_o$ or $\boldsymbol{t}_o$ is positive) by $\boldsymbol{n}_o$ or $\boldsymbol{t}_o$ units, as shown in Figure 1.2.3.

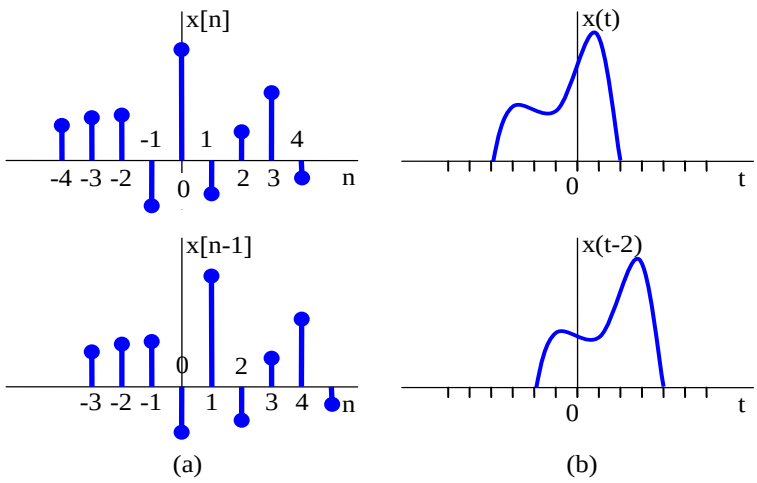

Fig. 1.2.3. (a) Discrete time and (b) Continuous time signals related by time shift.

Exploiting the same tape recorder example, this corresponds to reading the recording through a magnetic head placed at another position along the tape. Such delays are observed in sonar, radar, seismic signal processing and similar applications.

If the reflection yields the same signal i.e. if,

$$x(-t) = x(t) \tag{1.2.la}$$

or

$$x[-n] = x[n] \tag{1.2.1b}$$

then the signal is said to be "even". If, on the other hand, reflection yields the negative of the signal, i.e, if

$$x(-t) = -x(-t) \tag{l.2.2a}$$

or

$$x[-n] = -x[n] \tag{l.2.2b}$$

then this signal is said to be "odd". Every signal can be uniquely decomposed into an odd and an even part as

$$x(t) = x_o(t) + x_e(t). \tag{1.2.3}$$

where

$$x_o(t) = \frac{1}{2}[x(t) - x(-t)] \tag{1.2.4a}$$

and

$$x_e(t) = \frac{1}{2}[x(t) + x(-t)] \tag{1.2.4b}$$

Similar arguments are valid for discrete-time signals too.

A signal is said to be *periodic* if there exists a positive constant ***T*** such that

$$x(t+T) = x(t) \text{ for all } \boldsymbol{t}. \tag{1.2.5}$$

The fixed number ***T*** is called the *period* of $x(t)$. If $x(t)$ is periodic and ***T*** is a period, any integer multiple of ***T***, is also a period. .The smallest nonzero period ***T*** is called the *fundamental period*. If there exists no ***T*** for which (1.2.5) is satisfied, the signal $x(t)$ is said to be *aperiodic*.

A discrete-time signal (sequence) $x[n]$ is said to be *periodic* if there exists a positive integer ***N***, called the *period*, for which,

$$x[n+N] = x[N] \text{ for all } \boldsymbol{n} \tag{1.2.6}$$

The smallest nonzero period ***N*** is called the *fundamental period*. A signal which is not periodic for any ***N*** is *aperiodic*. Note once again that the period must be an integer, for a noninteger shift is not defined.

1.3. BASIC CONTINUOUS TIME SIGNALS

In this section some basic signal forms are introduced, for they appear frequently in nature and they serve as building blocks for many other more complex signals.

1.3.1. Complex- Exponential and Sinusoidal Signals

The continuous-time "complex exponential signal" is of the form

$$x(t) = Ce^{at} \tag{1.3.1}$$

where ***C*** and ***a*** are complex constants. Depending upon the .values of ***C*** and ***a***, the complex exponential takes different characteristics. If ***C*** and ***a*** are real, $x(t)$ is called a real exponential and shows two types of behavior shown in Figure l.3.l: if ***a*** >0, $x(t)$ is a growing exponential, and if ***a*** <0, it is a decaying exponential. The growing exponential represents phenomena such as, chain reactions of nuclear explosion, where as the decaying exponential describes phenomena like radioactive decay or discharge of a capacitor through a resistor. Note that if ***a***=0, then $x(t)$ is a constant.

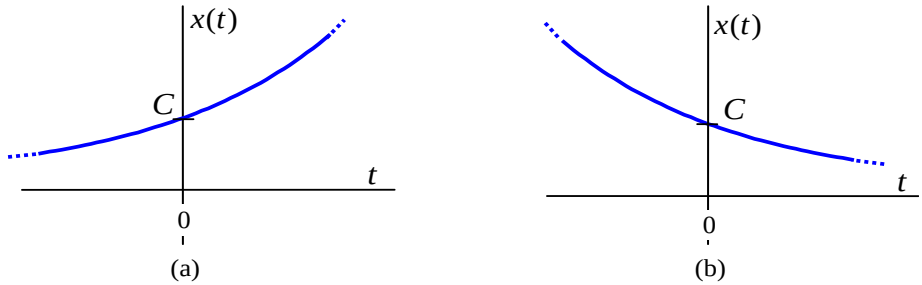

Figure 1.3.1. Continuous-time real exponential: (a) α>0, (b) α<0.

If ***a*** is purely imaginary, that is ,

$$x(t) = e^{j\omega_o t} \tag{1.3.2}$$

we have a periodic signal, the smallest or the *fundamental* period being:

$$T_o = 2\pi/|\omega_o| \tag{1.3.3}$$

Proof of these facts are left as an exercise for the reader. Note also that $e^{j\omega_o t}$ and $e^{-j\omega_o t}$ have the same period.

A signal closely related to the periodic complex exponential is the *sinusoidal signal*,

$$x(t) = Acos(\omega_0 t + \theta) = \frac{A}{2}e^{j\theta}e^{j\omega_0 t} + \frac{A}{2}e^{-j\theta}e^{-j\omega_0 t} \ . \tag{1.3.4}$$

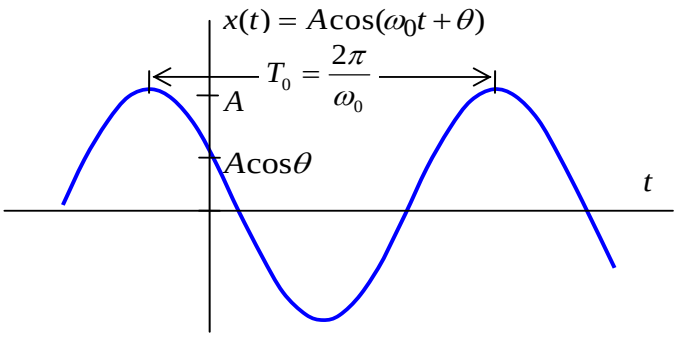

Fig.1.3.2. Continuous-time sinusoidal signal

This signal is shown in Figure 1.3.2.

Sets of periodic exponentials with fundamental frequencies (i.e. ω_0) that are. multiples of a single positive frequency, that is

$$x_k(t) = e^{jk\omega_0 t} \qquad k=0, \pm 1, \pm 2, \ldots\ldots \tag{1.3.5}$$

are said to be "harmonically related". Note that all these signals have a common (not necessarily fundamental) period $2\pi/\omega_0$.

The most general form of the complex exponential is obtained when

$$C = |C|e^{j\theta} \tag{1.3.6a}$$

and

$$\alpha = \varphi + j\omega_0 \ . \tag{1.3.6b}$$

Then

$$Ce^{\alpha t} = |C|e^{\varphi t}e^{j(\omega_0 t+\theta)}$$

$$= |C|e^{\varphi t}\cos(\omega_0 t + \theta) + j|C|e^{\varphi t}\sin(\omega_0 t + \theta). \qquad (1.3.7)$$

For $\varphi = 0$, both the real part and the imaginary part of the signal are sinusoidal. For, $\varphi > 0$ they correspond to a sinusoid multiplied by a growing exponential and for $\varphi < 0$, by a decaying exponential, as shown in Figure 1.3.3.

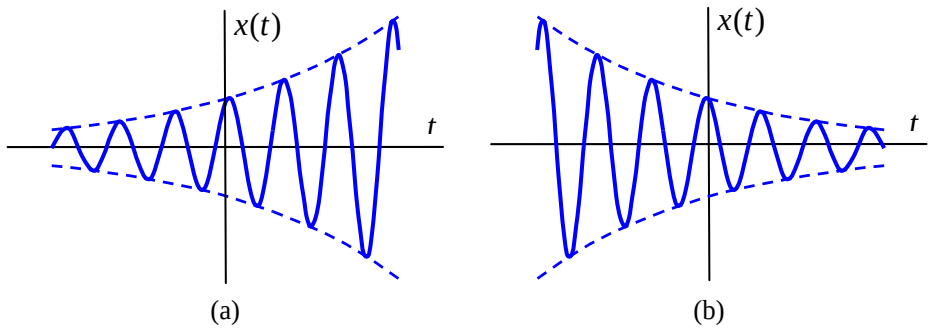

Fig. 1.3.3; (a) Growing sinusoidal signal (b) Decaying sinusoidal signal

The multiplying function $|C|e^{\varphi t}$, shown by dashed line is called the "envelope" of the sinusoid. Such signals are often called "damped sinusoids" and they arise in the response of RLC circuits and mechanical systems such as automotive suspension systems.

1.3.2. The Continuous-Time Unit Step and Unit Impulse

Another basic continuous -time signal is the "unit step function" $u(t)$, defined as

$$x(t) = \begin{cases} 0 & t < 0 \\ 1 & t > 0 \end{cases} \qquad (1.3.8)$$

which is shown in Figure 1.3.4.

Note that this function is discontinuous at t=0.

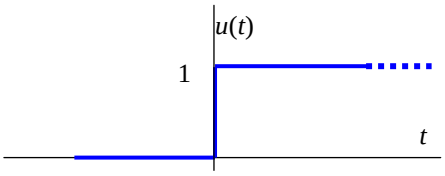

Fig. 1.3.4. The unit step function.

Another important basic signal is the "unit impulse function" $\delta(t)$ which is related to unit step function by

$$u(t) = \int_{-\infty}^{t} \delta(\tau)d\tau \qquad (1.3.9)$$

This suggests that

$$\delta(t) = \frac{du(t)}{dt} \qquad (1.3.10)$$

but there is obviously a formal difficulty there, due to the discontinuity. However if we define $u_\Delta(t)$ as in Figure l.3.5a so that

$$u(t) = \lim_{\Delta \to 0} u_\Delta(t) \qquad (1.3.11)$$

$\delta_\Delta(t)$, the time derivative of $u_\Delta(t)$ will be as depicted in Figure l.3.5b. As Δ approaches zero, $\delta_\Delta(t)$ becomes narrower and higher as it maintains its unit area. At the limit

$$\delta(t) = \lim_{\Delta \to 0} \delta_\Delta(t). \tag{1.3.12}$$

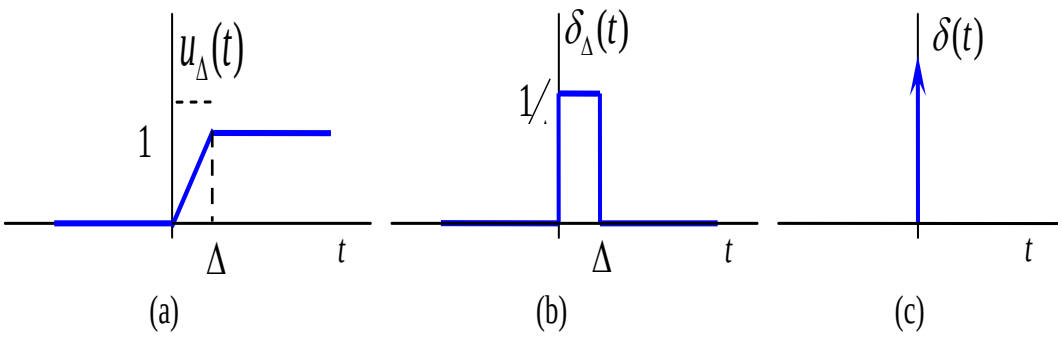

Fig. 1.3.5 (a) Continuous aproximation to unit step, (b) Derivative of $u_\Delta(t)$, (c) Unit impulse.

Note that $\delta(t)$ thus defined is not a formal function, it is defined only by its properties:

It is a signal of unit area, vanishing everywhere except at the origin,
It is the derivative of the unit step function.
The integral of the product $\varphi(t)\delta(t)$ equals $\varphi(0)$ for any $\varphi(t)$ continuous at the origin, if the interval of integration contains the origin, i.e.

$$\int_{t_1}^{t_2} \varphi(\tau)\delta(\tau)\, d\tau = \varphi(0) \quad \text{for } t_1 < 0 < t_2. \tag{1.3.13}$$

This is a very important property known as the "sifting property". Proof is left as an exercise.

1.4. BASIC DISCRETE TIME SIGNALS

1.4.1. The Discrete-time Unit Step and Unit Impulse Sequences

The counterpart of unit step function is the discrete time unit step sequence

$$u[n] = \begin{cases} 0 & n < 0 \\ 1 & n \geq 0 \end{cases} \tag{1.4.1}$$

and shown in Figure 1.4.1. Note that, this time, contrary to the continuous-time case, the value at the

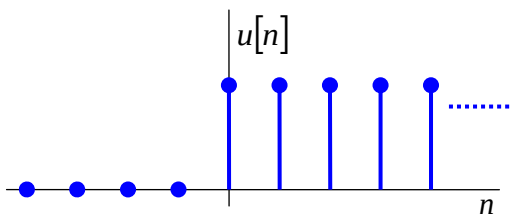

Fig. 1.4.1. Unit step sequence.

origin ***n***=0 is defined as "one".

The counterpart of the continuous-time unit impulse is the "unit sample" or the "unit impulse sequence" defined as

$$\delta[n] = \begin{cases} 0 & n \neq 0 \\ 1 & n = 0 \end{cases} \tag{1.4.2}$$

and shown in Figure 1.4.2.

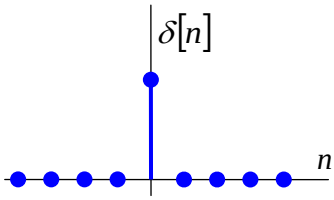

Fig. 1:4.2. Unit sample

The unit impulse sequence, like unit impulse function has some nice properties such as:

$$x[n]\delta[n] = x[0]\delta[n] \quad , \tag{1.4.3a}$$
$$\delta[n] = u[n] - u[n-1] \quad , \tag{1.4.3b}$$
$$u[n] = \sum_{i=-\infty}^{n} \delta[i] \quad , \tag{1.4.3c}$$
$$u[n] = \sum_{i=0}^{\infty} \delta[n-i] \quad , \tag{1.4.3d}$$
$$\sum_{i=k_1}^{k_2} x[i]\delta[i] = x[0] \qquad \text{if } k_1<0<k_2 \ . \tag{1.4.3e}$$

1.4.2. Discrete Time Complex Exponential and Sinusoidal Signals

The "complex exponential sequence" is defined by

$$x[n] = C\alpha^n = Ce^{\beta n} \tag{1.4.4}$$

with $\alpha = e^{\beta}$.

If C and α are real, we have one of the forms illustrated in Figure 1.4.3. Basically if $|\alpha| > 1$, the signal grows exponentially, while if $|\alpha| < 1$ we have a decaying exponential. If α is positive, all the samples have the same sign, but if α is negative successive samples have opposite signs.

Another important complex exponential signal is obtained by letting β to be purely imaginary, i.e.

$$x[n] = Ce^{j\Omega_0 n} \quad . \tag{1.4.5}$$

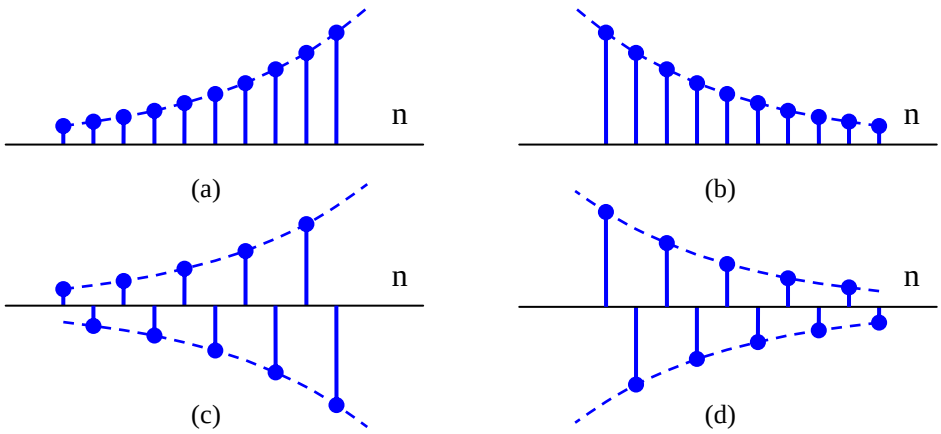

Fig.1.4.3. $x[n] = C\alpha^n$ (a) α>1, (b) 0<α<1, (c) -1<α<0, (d) α<-1

As in the continuous time case, this signal is closely related to the sinusoidal signal

$$x[n] = A\cos(\Omega_0 n + \theta) \quad . \tag{1.4.6}$$

A general complex exponential can be related to sinusoids as

$$C\alpha^n = |C||\alpha|^n\{\cos(\Omega_0 n + \theta) + j\sin(\Omega_0 n + \theta)\} \tag{1.4.7}$$

where

$$C = |C|e^{j\theta} \quad , \quad \alpha = |\alpha|e^{j\Omega_0} \quad . \tag{1.4.8}$$

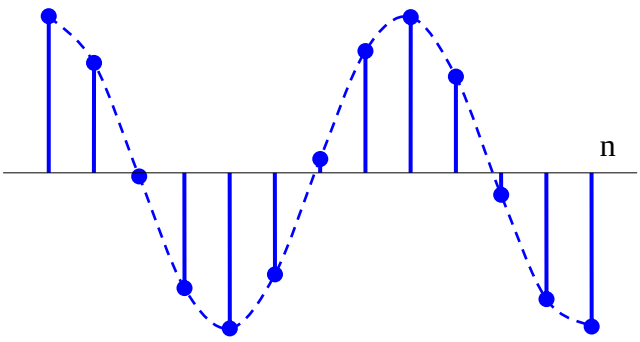

Fig. 1.4.4. A discrete time sinusoidal signal

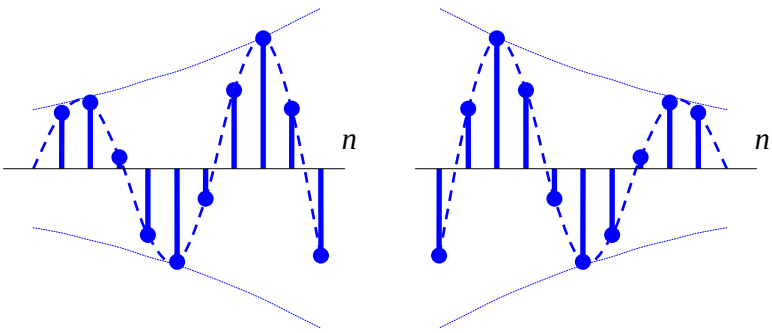

Fig. 1.4.5. (a) Growing, (b) decaying discrete time sinusoids.

Thus, the real and imaginary parts of a complex exponential are sinusoids multiplied by decaying or growing exponentials depending on whether $|\alpha| < 1$ or $|\alpha| > 1$ as depicted in Figure 1.4.5.

1.4.3. Periodicity Properties of Discrete Time Complex Exponentials

Recall that the continuous time signal $e^{j\omega_0 t}$ is periodic for any ω_0 and as the magnitude of ω_0 increases, rate of oscillation increases. But these properties are not valid for its discrete-time .counterpart $e^{j\Omega_0 n}$. The important difference arises from the fact that, although $e^{j\omega_0 t}$ are distinct signals for distinct ω_0 , $e^{j\Omega_0 n}$ with frequency Ω_0 is identical to that with frequency $\Omega_0 + 2\pi$.

For $e^{j\Omega_0 n}$ to be periodic we must find N such that

$$e^{j\Omega_0(n+N)} = e^{j\Omega_0 n}$$

or

$$e^{j\Omega_0 N} = 1 \, . \tag{1.4.9}$$

which implies that

$$\Omega_0 N = 2\pi m \tag{1.4.10}$$

where both ***m*** and ***N*** are integers. Therefore this signal is periodic only if $\Omega_0/2\pi$ is a rational number, To illustrate this point one can show that $x[n] = \cos n$ is aperiodic while $x[n] = \cos \pi n$ is periodic.

1.5. SYSTEMS

A "system" can be viewed as any process that results in the transformation of signals. It has an "input" signal and an "output" signal related by the system transformation. A continuous-time system transforms the input signal $x(t)$ into the output signal $y(t)$. This relation is shown by the notation

$$x(t) \rightarrow y(t). \tag{1.5.1}$$

Similarly, a discrete-time system transforms the input sequence $x[n]$ to an output sequence $y[n]$, i.e.

$$x[n] \rightarrow y[n]. \tag{1.5.2}$$

Such systems are shown pictorially as in Figure 1.5.1.

Figure 1.5.2(a) illustrates a "series" or "cascade" interconnection of two systems. Such diagrams are

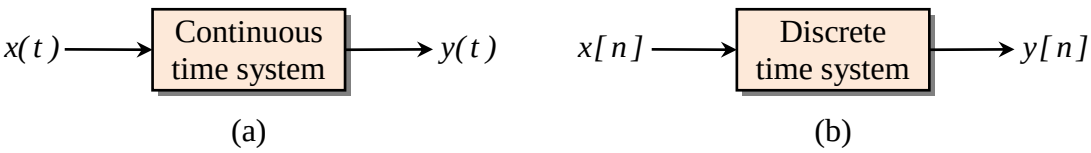

Fig.1.5.1. (a) Continuous-time system; (b) Discrete-time system.

referred to as block diagrams. Here the output of S1 is the input to S2. The overall system processes an input first by S1 and then by S2 and yields an output.

A "parallel" interconnection is shown in Figure 1.5.2(b). Here the same signal is applied to both S1 and S2; and the outputs are added to get a single output signal. One can define series or parallel interconnections of three or more systems. It is even possible to combine both cascade and parallel interconnections to obtain more complex interconnections such as the one inFigure 1.5.1(c).

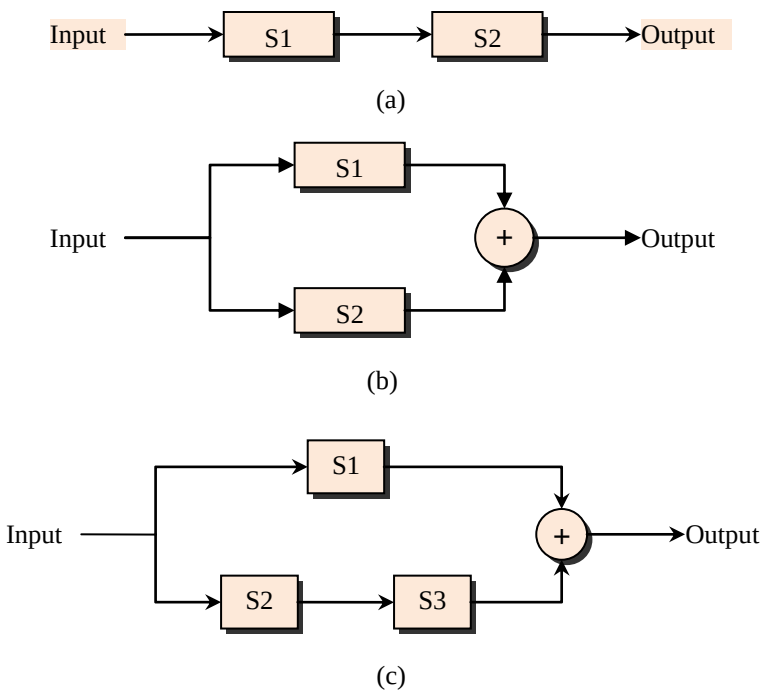

Fig.1.5.2. Interconnection of systems:

(a) Cascade; (b) parallel; (c) series/parallel

Another important interconnection is the "feedback" interconnection, illustrated in Figure 1.5.2. Here the output of S1 is processed through S2 and fed back and added to the external input to produce the input to S1.

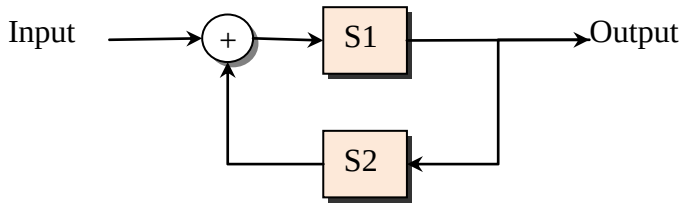

Fig.1.3.2. Feedback interconnection.

1.6. PROPERTIES OF SYSTEMS.

In this section we introduce and discuss some basic properties of systems.

1.6.1. Memory

A system is said to be "memoryless" or "instantaneous" if the output at any instant depends only. on the value of the input at that particular instant. For example if a resistor is viewed as a system with $x(t)$ the current through and $y(t)$ the voltage across, it will be a memoryless system since

$$y(t) = Rx(t) \quad . \tag{1.6.1}$$

But the same argument is not valid for a capacitor since

$$y(t) = \frac{1}{C}\int_{-\infty}^{t} x(\tau)\, d\tau \tag{1.6.2}$$

depends on the values of the input prior to $\boldsymbol{t}$. A system which is not memoryless is a "system with memory". The capacitor mentioned above, the continuous time system defined by (pure delay)

$$y(t) = x(t-1) \tag{1.6.3}$$

or the discrete time system

$$y[n] = \sum_{k=-\infty}^{n} x[k] \tag{1.6.4}$$

are systems with memory.

1.6.2. Causality:

A system is said to be "causal", or "non anticipatory" or "realizable" if, it does not respond before being stimulated. In other words output at any time depends only on values of the input at the present time and in the past. Note that all the memoryless systems are causal. The capacitor example cited earlier is a causal system. In fact all physical systems are causal unless ambiguously modeled (Hence, the term "realizable"). But,

$$y[n] = x[n] - x[n+1] \tag{1.6.5a}$$

or

$$y(t) = x(t+1) \tag{1.6.5b}$$

are non-causal while

$$y[n] = x[n] - x[n-1] \tag{1.6.6a}$$

$$y(t) = x(t-1) \tag{1.6.6b}$$

are causal.

Note that for the analysis of physical systems in real time, causality is an essential physical prerequisite. However there are practical applications for non-causal systems too. Off-line computer processing or applications where the independent variable is not time allow for non causal system representations.

1.6.3. Stability

A "stable" system is one in which small inputs lead to outputs that do not diverge. For example a ball resting at the base of a valley as shown in Figure l.6.l(a) constitute a stable system since small disturbances in horizontal position (input) will lead only to small perturbations in the vertical position (output). However the ball resting at the top of a hill as shown in Figure l.6.1(b) will constitute

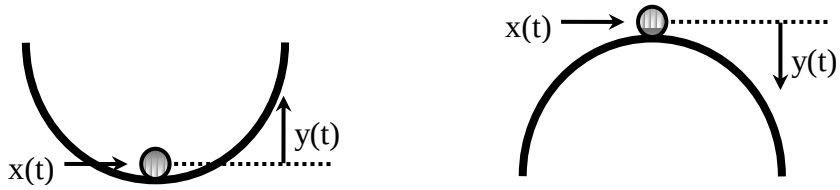

Fig. 1.6.1. Examples of (a) stable, (b) unstable systems.

an "unstable" system, because a very small perturbation from the rest position will cause the ball to roll down the hill and deviate from the equilibrium indefinitely.

Basically if the input to a stable system is bounded, then the output must also be bounded. For example the system described by

$$y[n] = \sum_{k=-M}^{M} x[n-k] \tag{1.6.7}$$

is stable since if $x[n]$ is less than some ***B*** for all ***n***, then $y[n]$ is less than ***(2M+1)B*** for all ***n***. On the, other hand

$$y[n] = \sum_{k=-n}^{n} x[k] \tag{1.6.8}$$

represents an unstable system, since for $x[n]=1$ the the magnitude of the output

$$y[n] = \left|\sum_{k=-n}^{n} 1\right| = 2|n| + 1 \tag{1.6.9}$$

will grow without bound.

1.6.4. Time Invariance

A system is "time-invariant" if a time shift at the input causes a time shift at the output, by the same amount. Or mathematically a system is time invariant if

$$x(t) \rightarrow y(t) \tag{l.6.10.a}$$

implies

$$x(t-t_0) \rightarrow y(t-t_0) \quad \text{for all } \mathbf{t_0}. \tag{l.6.10.b}$$

As an example consider the system described by

$$y(t) = \sin[x(t)]. \tag{1.6.11}$$

To check time invariance, let $x_1(t)$ be any input to this system with the response

$$x_1(t) \rightarrow y_1(t) = \sin[x_1(t)]. \tag{1.6.12}$$

Now, when the input is shifted, the new input

$$x_2(t) = x_1(t-t_0) \tag{1.6.13}$$

will cause a response

$$x_2(t) \rightarrow y_2(t) = \sin[x_2(t)] = \sin[x_1(t - t_0)] \tag{1.6.14}$$

which is obviously equal to $y_1(t\text{-}t_0)$. Hence the system is time invariant.

As a second example, consider

$$y[n] = nx[n] \tag{1.6.15}$$

and consider the responses to $x_1[n]$ and $x_2[n]=x_1[n\text{-}n_0]$:

$$x_1[n] \rightarrow y_1[n] = nx_1[n];$$

$$x_2[n] \rightarrow y_2[n] = nx_2[n] = nx_1[n - n_0]\ . \qquad '(1.6.16)$$

However, if we shift $y_1[n]$, we obtain

$$y_1[n - n_0] = |nx[n]|_{n \rightarrow n - n_0} = (n - n_0)x_1[n - n_0] \tag{1.6.17}$$

and therefore the system is "time-varying".

1.6.5. Linearity

A "linear" system, in both continuous and discrete time, is one that has the property of superposition: If an input is the weighted sum of several signals, then the output is the same weighted sum of the responses to each of these signals. Mathematically let

$$\begin{matrix} x_1(t) \rightarrow y_1(t) \\ x_2(t) \rightarrow y_2(t) \end{matrix}. \tag{1.6.18}$$

Then the system is linear if

$$x_1(t) + x_2(t) \rightarrow y_1(t) + y_2(t) \tag{1.6.19}$$

and

$$a.x_1(t) \rightarrow a.y_1(t) \tag{1.6.20}$$

for any $\boldsymbol{x_1}(t)$, $\boldsymbol{x_2}(t)$ any complex constant $\boldsymbol{a}$.

The first one of these properties is called the "additivity" property and the second is known as the "homogeneity" property. A system which is both additive and homogeneous is said to be linear. The two properties can be combined into a single statement as

$$ax_1(t) + bx_2(t) \rightarrow ay_1(t) + by_2(t) \tag{1.6.21}$$

for any $\boldsymbol{x_1}(t)$, $\boldsymbol{x_2}(t)$, $\boldsymbol{a}$ and $\boldsymbol{b}$.

The discussions presented above for continuous-time systems are equally valid for discrete time systems.

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(*) Of primary importance.

PROBLEMS.

Problem 1.

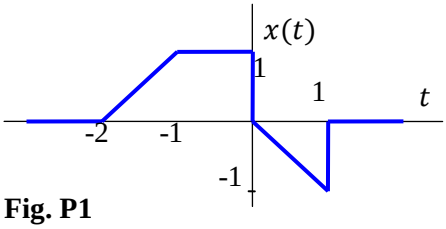

Fig. P1

A continuous-time signal $x(t)$ is shown in the figure. Sketch and carefully label each of the following signals:

i) $x(t-2)$,
ii) $x(2-t)$,
iii) $x(2(t-2))$,
iv) $x(-2(t+1/2))$.

Problem 2.

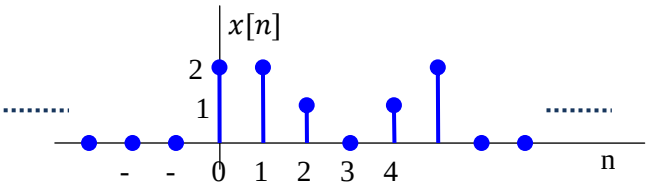

Fig. P2

For the sequence depicted in the figure, carefully sketch and label the following sequences:

i) $x[n-2]$,
ii) $x[3-n]$,
iii) $x[2-2n]$,
iv) $x[1-n]u[n]$,
v) $x[n]+x[-n]$.

Problem 3.

A continuous-time dynamic system is known to be time-invariant. Two tests are run-on this system and the input-output pairs shown in Figure P1 are obtained. Discuss the linearity of this system.

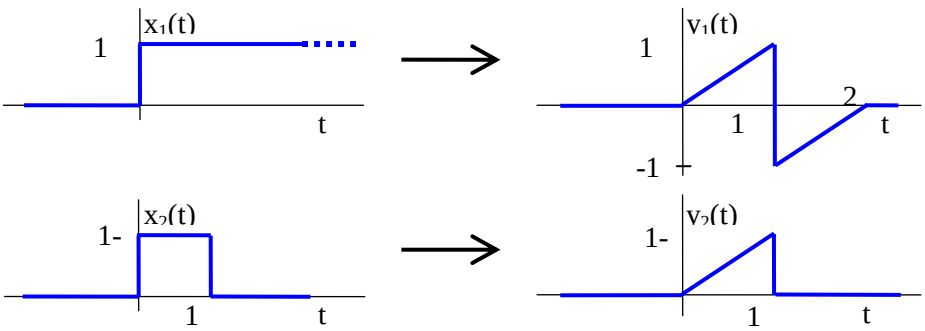

Fig.P3

Problem 4.

Following tests are run on a discrete-time dynamical system, with, the system initially at rest:

$$x_1[n] = \begin{cases} 1 & n = 0 \\ -1 & n = 1 \\ 0 & otherwise \end{cases} \rightarrow y_1[n] = \begin{cases} 1 & n = 1 \\ 1 & n = 2 \\ 0 & otherwise \end{cases}$$

$$x_2[n] = \begin{cases} 1 & n = 1 \\ -1 & n = 2 \\ 0 & otherwise \end{cases} \rightarrow y_2[n] = \begin{cases} 1 & n = 1 \; or \; 3 \\ 0 & otherwise \end{cases}$$

$$x_3[n] = \begin{cases} 1 & n = 0 \; or \; 1 \\ -2 & n = 2 \\ 0 & otherwise \end{cases} \rightarrow y_3[n] = \begin{cases} 2 & n = 1, 3 \; or \; 4 \\ 0 & otherwise \end{cases}$$

Determine if this system is

- **i)** Linear?
- **ii)** Time (shift)-invariant?

Problem 5.

For the system given in figure P5 where $x(t)$ is the input and $y(t)$ is the output, determine if the

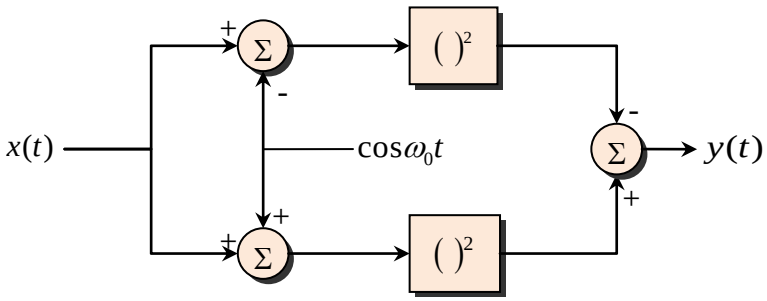

Fig.P5

relationship between $x(t)$ and $y(t)$ is

- **i)** 1inear,
- **ii)** time-invariant.

Problem 6.

In the following a general system with an input $x(t)$ or $x[n]$ and an output $y(t)$ or $y[n]$ is considered. Determine if each statement is true or false. In each case fully justify (i.e. prove or disprove) your answer.

- **i)** The system is not linear if $a.x(t) \rightarrow b.y(t)$, a not equal to b.
- **ii)** The system is linear if $k.x(t) \rightarrow k.y(t)$ for all k.
- **iii)** $y[n] = x[n^2]$ is causal.
- **iv)** Same as (iii) with $y[n] = x[n^2]u[-n]$.
- **v)** Same as (iii). with $y[n] = x[n^2]u[n]$.
- **vi)** $y[n] = x[n^2]$ is nonlinear.
- **vii)** $y[n] = x[n]u[-n]$ is causal.
- **viii)** $y(t) = x(-t)$ is time-invariant.
- **ix)** If the system is linear, then $k(t).x(t) \rightarrow k(t).y(t)$ for all $k(t)$.
- **x)** If the response to $x(t) = Ke^{j\omega t}$ is $y(t) = K\omega^2 e^{j\omega t}$ for all ω and K, then the system is linear.

xi) If the response to $x(t) = Ke^{j\omega t}$ is $y(t) = Ke^{j\omega t}$ for all ω and K, then the system is nonlinear.

xii) The series (cascade) connection of an unstable LTI system with another LTI system is necessarily unstable.

xiii) The system is time-invariant i $x(t - t_0) \rightarrow y(t)$ for all t0.

xiv) Cascade of a causal system with a non-causal system is always non-causal.

xv) The combination of (xiv) is always causal.

xvi) If the system is time-invariant, then $x(t - t_0) \rightarrow y(t)$, for all t0.

xvii) If the system is linear, time-invariant and, $|h(t)| < \infty$ for all t, then the system is stable.

xviii) If the system is linear, time-invariant and $|h(t)| = 0$ for $|t| < t_0$, then the system is causal.

Problem 7.

Determine if the system $y(t) = x(2t)u(t)$ is

i) time-invariant,

ii) causal?

Problem 8.

Consider the continuous-time system characterized by the input-output relationship:

$$y(t) = \begin{cases} x(at + b) & \text{for } t \neq 0 \\ K & \text{for } t = 0 \end{cases}.$$

Determine the values of the parameters a, b and K for this system to be

i) Memoryless,
ii) Causal,
iii) Stable,
iv) Linear,
v) Time invariant.

Prove your results.

Problem 9.

The input-output relation of a system $\{x(t) \longrightarrow y(t)\}$ is:

$$y(t) = \begin{cases} 3x(t) & x(t-2) \geq 0 \\ 2e^{t}x(t) & x(t-2) < 0 \end{cases}.$$

Determine if this system is

i) memoryless,
ii) causal,
iii) stable,
iv) linear,
v) time-invariant.

Problem 10.

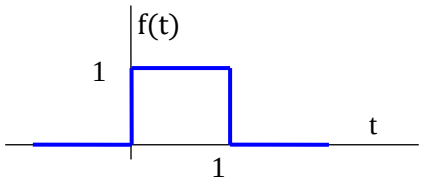

Fig. P 10

A system is described by the input-output relationship

$$x(t) \longrightarrow y(t) = \begin{cases} \int_0^{\infty} x(\tau) f(t+\tau) d\tau & t \geq 0 \\ 0 & t < 0 \end{cases}$$

where the function $f(t)$ is as given in the figure. Determine with full justification, whether this system is

- **i)** Linear;
- **ii)** Time-invariant;
- **iii)** Causal;
- **iv)** Stable.

Problem 11.

a) Consider a system **S1**, whose input-output relationship is given by $x[n] \longrightarrow y[n] = x[-n]$.

Determine whether or not the given system is

- **i)** memoryless,
- **ii)** linear,
- **iii)** time-invariant,
- **iv)** causal,
- **v)** stable.

In each case justify your answer.

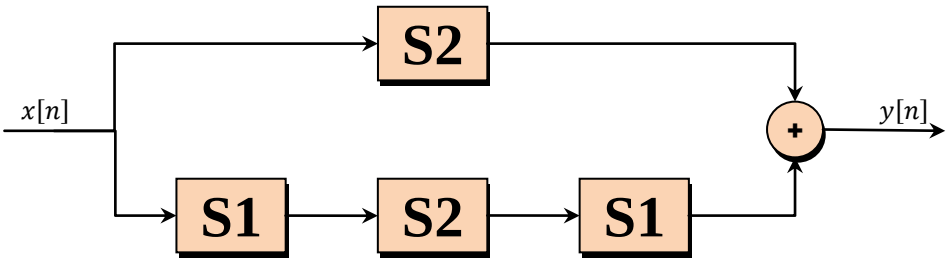

Fig. P11

b) Let **S2** be a causal, stable, LTI system with memory, having an impulse response h[n]. Obtain the input-output relationship for the system shown in the figure and determine whether or not this new system is

- **i)** memoryless,
- **ii)** linear,
- **iii)** time-invariant,
- **iv)** causal,
- **v)** stable.

In each case justify your answer.

Problem 12.

Determine whether each one of the following statements is TRUE or FALSE and fully justify your answer (i.e. prove if true and disprove or give a counter example if false.):

i) If a system described by the input/output relationship $x(t) \longrightarrow y(t)$ is linear, then $x(t+T) \longrightarrow y(t) + y(T)$.

ii) If a system is causal, its output is zero for all t, when the input is identically zero for all t.

iii) A system, described by the input/output relationship $x(t) \longrightarrow y(t)$ is time invariant if

$$x(t - t_0) \longrightarrow y(t) \ .$$

iv) A general discrete-time dynamic system $x[n] \longrightarrow y[n]$ is stable if and only if $|y[n]|$ is finite for all input sequences satisfying $|x[n]| = \infty$.

v) The system described by $x[n] \longrightarrow y[n] = x^*[n]$ is linear.

vi) The system described by $y[n] = x[n-1].h[n]$ is time- invariant.

vii) The system described by $x[n] \longrightarrow y[n] = x[n-1]f[n]$ is time- invariant, for an arbitrary sequence $f[n]$.

viii) $f(t)\delta(t - t_0) = f(t_0)$.

ix) Let a discrete-time sequence x[n] be obtained by sampling a continuous-time signal $x(t)$ according to $x[n] = x(nT)$ for n=0, ±1, ±2, ….. If $x[n]$is periodic, then $x(t)$ is also periodic.

Problem 13.

A linear, time-invariant system yields the output $y_1(t)$ when it is excited by the signal $x_1(t)$, both shown in figure (a). Determine and sketch the response to $x_2(t)$, shown in figure (b).

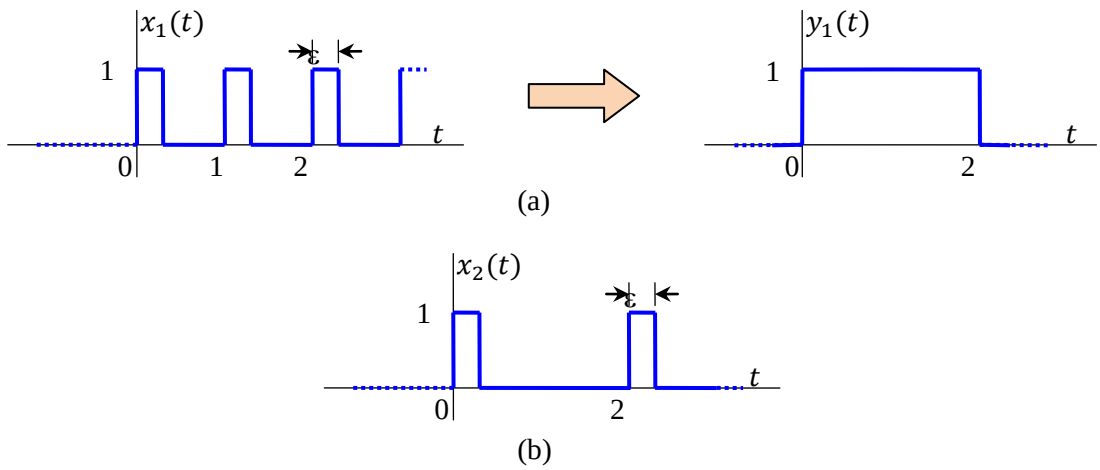

Fig. P13

Problem 14.

x(t) is a periodic signal consisting of sinusoidal pieces as shown in the figure.

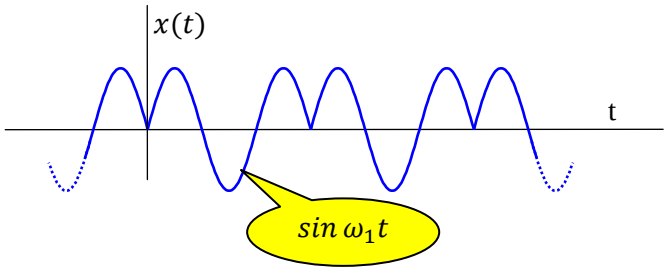

Fig. P14

a) What is the fundamental frequency of $x(t)$ in terms of ω_1.
b) Consider a discrete-time sequence $x[n]$ obtained by sampling $x(t)$ at $t = nT_s$, where n=0, ±1, ±2, ±3. Determine the condition on T_s in terms of ω_1 so that $x[n]$ is a periodic sequence.

Problem 15.

Consider the continuous-time system characterized by the input-output relationship:

$$x(t) \longrightarrow y(t) = \begin{cases} x(t) & \text{for } t \neq 0 \\ K & \text{for } t = 0 \end{cases}.$$

Determine the value of the parameter K for this system to be

i) Linear,
ii) Time invariant

Prove your results.